

Pulverised fuel firing:- This is most modern and efficient method of boiler firing. Solid fuels can be used in a powdered form and burn like oil and gases. The coal is first dried usually by the gases, ground to a fine powder in a pulverised mill and then projected into the combustion chamber by means of a current of hot air. A volume of preheated air to make up the necessary amount for combustion is blown in separately, and the resulting turbulence in the high pressure combustion chamber helps in thorough combustion of the fuel. The air used to dry the coal and convey the powdered fuel to the furnace is called the primary air and the air blown separately to complete the combustion is called the secondary air.

Assignment 2.

Describe with sketches the firing systems for —
Liquid and gaseous fuels.

Example 1.

A boiler evaporates 3.6 kg of water per kg of coal into dry saturated steam at 10 bar. The temperature of feed water is 32°C . Find the equivalent evaporation from and at 100°C so well as the factor of evaporation.

Solution

Given: mass of water actually evaporated (m_e) = 3.6 kg
Pressure (P) = 10 bar
Temperature of feed water (t_f) = 32°C

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Equivalent evaporation "from and at 100°C (E) = ?
 Factor of evaporation (F_e) = ?

$$E = \frac{m_e (h - h_{f1})}{2257}$$

h_{f1} from steam tables at feed water temperature of $32^{\circ}\text{C} = 134\text{ kJ/kg}$
 and $h = h_g$ from steam tables at pressure 10 bar = 2778 kJ/kg

$$\Rightarrow E = \frac{3.6 (2778 - 134)}{2257} = \frac{9518.4}{2257}$$

$$\text{and } F_e = \frac{h - h_{f1}}{2257} = \frac{2778 - 134}{2257} = \frac{2664}{2257}$$

$$= 4.2\text{ kg/kg of coal}$$

$$= 1.17$$

Example 2:

The following observations were made in a boiler trial:

Coal used 250 kg of calorific value 29800 kJ/kg , water evaporated 2000 kg , steam pressure 11.5 bar , dryness fraction of steam 0.95 and feed water temperature 34°C . Calculate the equivalent evaporation "from and at 100°C per kg of coal and the efficiency of the boiler.

Solution

Given: mass of fuel (m_f) = 250 kg ;

Calorific value of fuel (C) = 29800 kJ/kg

mass of water evaporated into steam (m_s) = 2000 kg

(13)

Steam pressure (P) = 11.5 bar

dryness fraction (x) = 0.95

Feed water temp. (t_f) = 34°C

Equivalent evaporation (E) = ?

Boiler efficiency (η) = ?

Equivalent evaporation from and at 100°C is :

$$E = \frac{m_e (h - h_f)}{2257}$$

h_{fg} from steam table at feed water temperature of $34^\circ\text{C} = 142.4 \text{ kJ/kg}$

h_{fg} from steam table at pressure 11.5 bar = 1993 kJ/kg

and h_f from steam table at pressure 11.5 bar = 789.5 kJ/kg

Entropy (h) = $h_f + x h_{fg}$

$$\begin{aligned} &= 789.5 + 0.95(1993) \\ &= 2682.9 \text{ kJ/kg} \end{aligned}$$

$$m_e = \frac{m_s}{m_f} = \frac{2000}{250} = 8 \text{ kg/kg of coal}$$

$$E = \frac{8(2682.9 - 142.4)}{2257}$$

$$= \frac{20324}{2257} = 9 \text{ kg/kg of coal}$$

Efficiency of the boiler is :

$$\eta = \frac{m_e (h - h_f)}{C} = \frac{8(2682.9 - 142.4)}{29800} = \frac{20324}{29800}$$

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$$= 0.682$$

$$= 68.2\%$$

Example 3:

A Lancashire boiler generates 2400 kg of dry steam per hour at a pressure of 11 bar. The grate area is 3 m^2 and 90 kg of coal is burnt per m^2 of grate area per hour. The calorific value of the coal is 33180 kJ/kg and the temperature of feed water is 17.5°C . Determine:

(a) Actual evaporation

(b) Equivalent evaporation

(c) Efficiency of the boiler.

from and at 100°C , and

Solution

Given: Mass of steam (m_s) = 2400 kg/h; pressure = 11 bar; grate area = 3 m^2 ; Coal burnt = 90 kg/ m^2/h ; Calorific value of fuel (C) = 33180 kJ/kg; temperature of feed water (t_f) = 17.5°C

(a) Actual evaporation per kg of coal ($m_e = \frac{m_s}{m_f}$)

$$m_f = 90 \times 3 = 270 \text{ kg/h}$$

$$\therefore \text{Actual evaporation per kg of coal (} m_e \text{)} = \frac{m_s}{m_f} = \frac{2400}{270}$$

$$= \underline{\underline{8.89 \text{ kg/h}}}$$

(b) Equivalent evaporation 'from and at 100°C ' (E):

$$E = \frac{m_e(h - h_f)}{2257}$$

h_f from steam tables at feed water temperature of $17.5^\circ\text{C} = 73.4 \text{ kJ/kg}$ and h from steam tables at 11 bar, $h_g @ 11 \text{ bar} = 2781 \text{ kJ/kg}$

$$\therefore E = \frac{8.89(2781 - 73.4)}{2257} = \frac{24070.564}{2257} = \underline{\underline{10.66 \text{ kg/h}}}$$

(14)

$$C) \text{ Efficiency of the boiler } (\eta) = \frac{m_e (h - h_{fi})}{C}$$

$$= \frac{8.89(2781 - 73.4)}{33180}$$

$$= \frac{24070.564}{33180}$$

$$= 0.725 = 72.5\%$$

Example 4:

A Steam power station has an overall efficiency of 20% and 0.6 kg of coal is burnt per kWh of electrical energy generated. Calculate the calorific value of fuel.

Solution

Let x cal/kg be the calorific value of fuel.

Heat produced by 0.6 kg of coal = $0.6x$ kcal

Heat equivalent of 1 kWh = 860 kcal

$$\text{From Overall efficiency } (\eta_{\text{overall}}) = \frac{\text{Electrical Output in heat units}}{\text{Heat of Combustion}}$$

$$0.2 = \frac{860}{0.6x}$$

$$x = \frac{860}{0.2 \times 0.6} = 7166.67 \text{ kcal/kg}$$

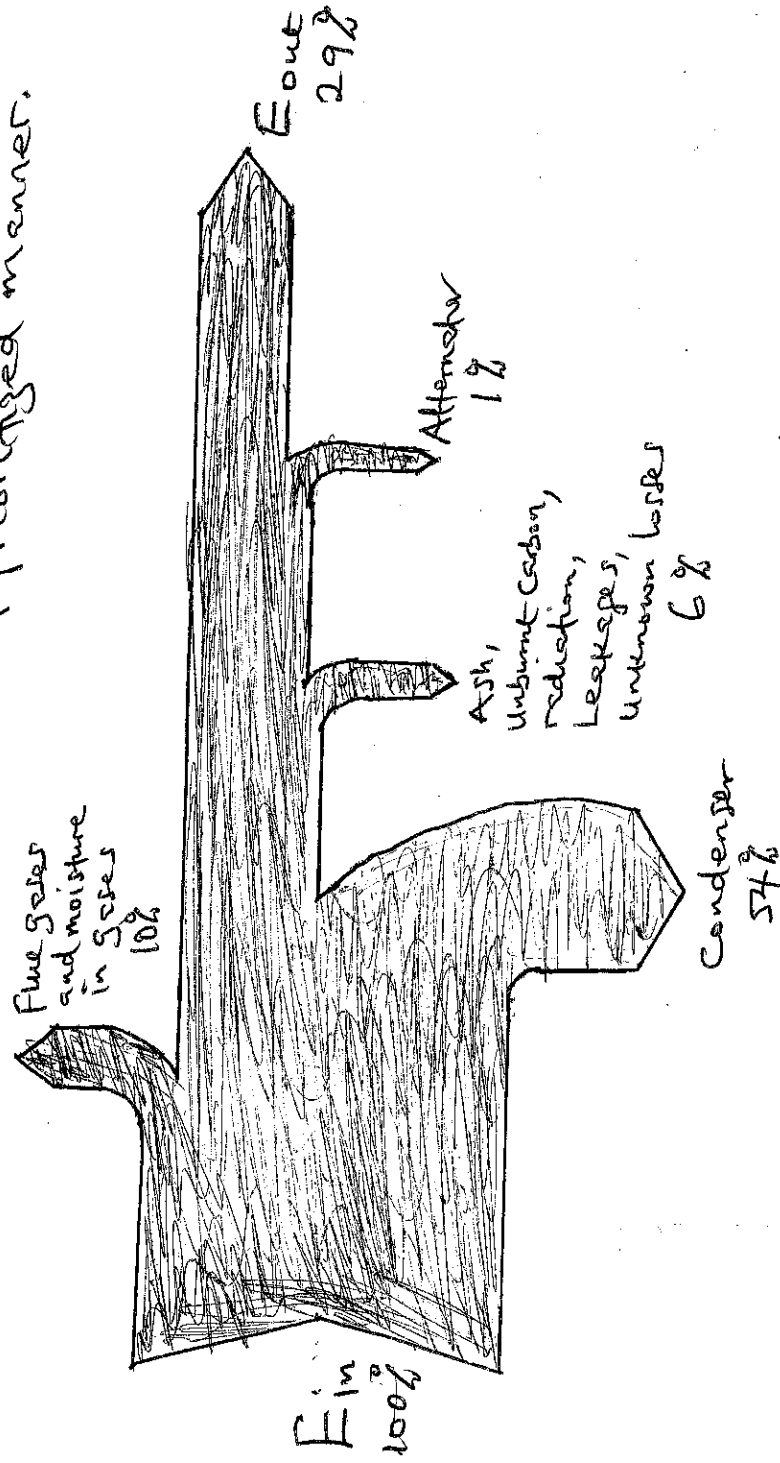
$$= 8.33 \text{ kWh/kg}$$

$$\therefore \text{ Calorific value of fuel} = 8.33 \text{ kWh/kg}$$

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The Sankey Diagram :- This is a flow diagram that represents an entire input and output energy flow in any equipment or system such as boiler, fired heaters or furnaces after carrying out energy balance calculation.

Sankey diagrams feature directed arrows that have a width proportional to the flow quantity. It visually represents various outputs and losses so that energy managers can focus on finding improvements in a prioritized manner.



Sankey diagram.

Pipe Sizing

Two principal factors determine pipe sizing in a steam system:

1. The initial pressure at the boiler and the allowable pressure drop of the total system. The total pressure drop in the system should not exceed 20% of the total maximum pressure at the boiler. This includes all drops — line loss, elbows, valves etc.
2. Steam velocity: Erosion and noise increase with velocity. Another consideration is future expansion.

Typical Pipe material for Steam Systems

Pipes for Steam Systems are commonly manufactured from Carbon Steel to ASME B 16.9 A106. The same material may be used for Condensate lines, although Copper tubing is preferred in some industries.

For high temperature Superheated mains, additional alloying elements, such as Chromium and molybdenum are included to improve tensile strength and creep resistance at high temperatures. Typically, pipes are supplied in 6m lengths.

Piping Layout

Piping layout is the representation of the arrangement of the overall piping system. It shows the piping, equipment nozzles, structure, piping supports, instruments' tag etc from the starts to the ends.

Best Practices for Industrial Steam Boiler Maintenance

Industrial boiler maintenance is critical for ensuring the optimal efficiency and safety of a facility. The best practices to keep boiler system running at its best include:

1. Monitor water quality and maintain proper water treatment processes: The hot water temperature inside the boiler system must receive chemical treatment to prevent scaling and damage to the boiler. Regular maintenance and chemical treatment of boiler feedwater prevent non condensable gases from getting inside the boiler and working havoc through "oxygen pitting".
2. Schedule regular steam boiler inspections: This helps to identify any weaknesses in the boiler's heating system. Inspections can identify areas of corrosion as well as threats of insulation failure and tube failure caused by oxygen pitting.
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3. Choose a Steam System with advanced boiler monitoring. Boilers should be monitored by trained maintenance personnel who know what to look for in a boiler system.
4. Choose a boiler system with the highest safety rating. In some boilers, the boilers will not start if the operator attempts to bypass critical safety measures.
5. Attention should be paid to the boiler system lifecycle. Typical lifecycle for a boiler is 15-25 years, the better it is maintained the longer we expect the boiler to last under normal conditions.

Hydro-electric power plant

Hydroelectric power plant is the plant that utilizes the potential energy of water at a high level for the generation of electrical energy.

This power plant is generally situated in hilly areas where dams can be built conveniently and large water reservoirs can be obtained. In a hydro-electric power station, water head is created by constructing a dam across a river or lake. From the dam, water is led to a water turbine. The water turbine captures the energy in the falling water and changes the hydraulic energy into mechanical energy at the turbine shaft. The turbine drives the alternator which converts mechanical energy into electrical energy.

Selection of site for hydroelectric power plants.

The following factors need to be considered in the selection of site for hydroelectric power plants:

1. Availability of water: Such plants should be constructed at a place where adequate water is available at a good head. Since the primary requirement of such power plant is the availability of large quantity of water.
2. Water Storage: Since there is wide variations in water supply from a river or canal during the year, it is necessary

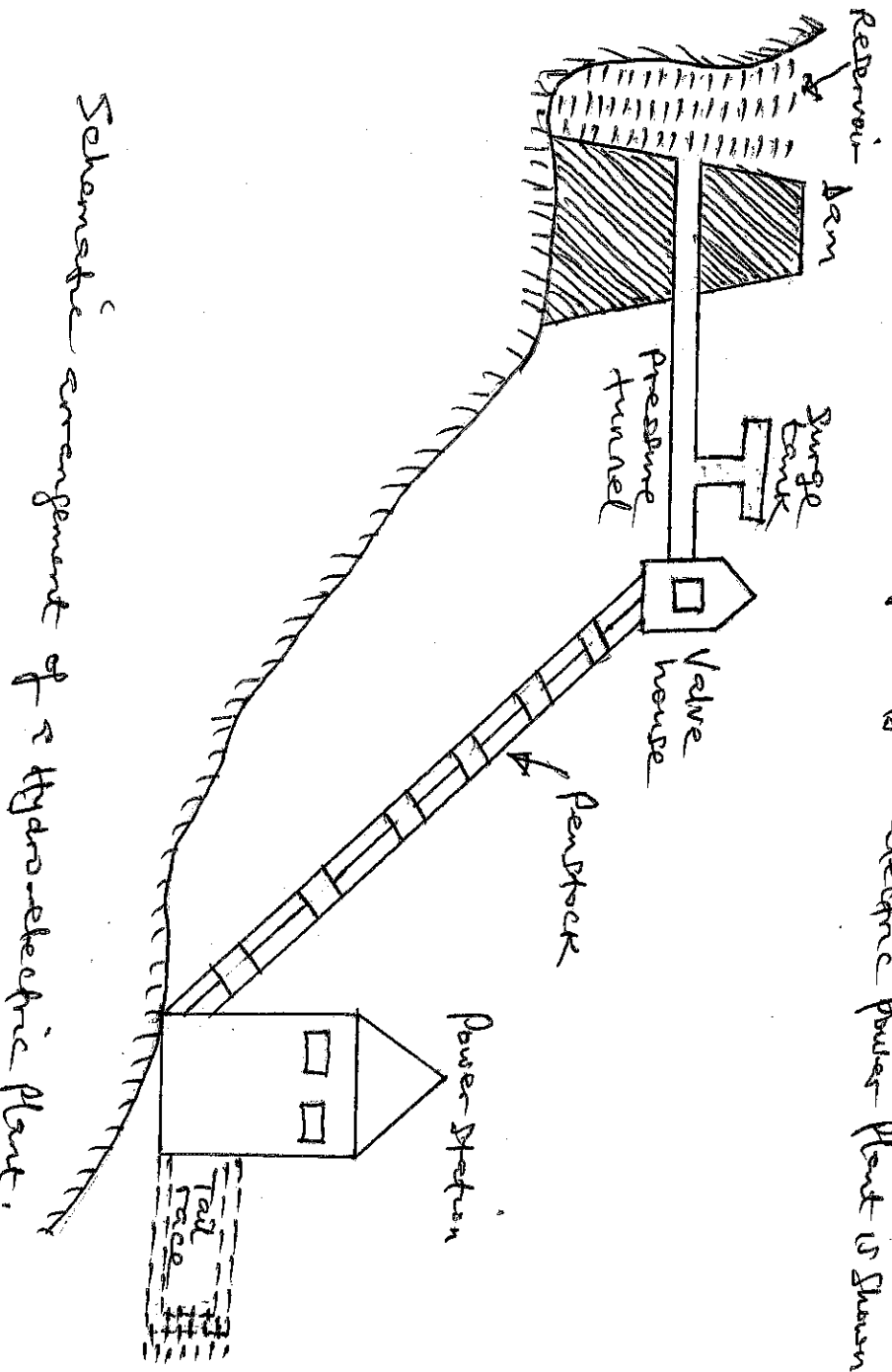
to store water by constructing a dam in order to ensure the continuous supply of water to the plant and consistent generation of power throughout the year.

3. Cost and type of land:- The land for the construction of the plant should be available at a reasonable price and its bearing capacity should be adequate to withstand the weight of the heavy equipment to be installed.

4. Accessibility of the site:- The site should be one that can be easily accessible by road or rail, so that necessary equipment and machinery can be easily transported.

5. Distance from load centre:- This plant is usually situated at a distance from the load centre. For economical transmission of electric power, the routes and distances need serious considerations.

The schematic arrangement of a hydro-electric power plant is shown below.



Schematic arrangement of a Hydro-electric Plant.
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Elements of hydro-electric power plant.

Some of the basic elements that make up the hydroelectric power plant are:

1. Reservoir :- It serves to store water during excess flow periods (i.e., rainy season) and supply it during lean flow periods (i.e., dry season) and thus helps in adequate supply of water to the turbines according to the load on the plant. It can be either a natural reservoir or artificial reservoir. A natural reservoir is a lake in high mountains and artificial reservoir is made by constructing a dam across the river.
2. Dam :- A dam aids in creating a water head and also provides a barrier which stores water and the facility of diversion into conduits. They are built of concrete or stone masonry, earth or rock fill. A masonry dam may be built in a narrow canyon. An earth dam may be best suited for a wide valley.
3. Spillway :- This is constructed to act as a safety valve. It discharges the overflow water to the down stream side when the reservoir is full.
4. Headworks :- The headworks consists of the diversion structures at the intake for diverting floating debris and sediments, and also valves for controlling the flow of water to the turbine. The flow of water into and through headworks should be as smooth as possible to avoid head loss and cavitation. It is necessary to avoid sharp corners and abrupt contractions or enlargements.
5. Surge tank :- It is a small reservoir or tank (open at the top) in which water level rises or falls to reduce the pressure swings in the conduit. A reduction in load on the generator causes the governor to close the turbine gates and thus creates an increased pressure in the penstock, and may result in water hammer phenomenon that may make the penstock to burst. The surge tank is there to prevent the water hammer pressure.

6. Penstocks:- They are open or closed conduits which carry water to the turbines. They are generally made of reinforced concrete or steel. Concrete penstocks are suitable for low head (< 30m) as greater pressure causes rapid deterioration of concrete. The steel penstocks can be designed for any head; the thickness of the penstock increases with the head or working pressure.

7. Water turbines:- They are used to convert the energy of falling water into mechanical energy.

8. Power house:- This is a building that houses the turbines, generators and other accessories.

9. Tail race:- The water after doing its useful work in the turbine is discharged to the tail race which may lead it to the same stream or to another. It should be such that water has a free exit with no interruption.

Advantages and Disadvantages of Hydroelectric Power Plant

Advantages:

1. Requires no fuel as water is the source of energy.
2. It is neat/clean as no smoke or ash is produced.
3. It is comparatively simple in construction and requires less maintenance.
4. Needs low running/operational cost.
5. It is robust and has a longer life. (Efficiency doesn't fall with age).
6. Also serves in irrigation and flood control purposes.
7. Highly skilled personnels are needed only at the time of construction for operation, experienced person(s) may do the job well.
8. It can start in few minutes i.e. does not require long starting time like that of a steam power station.

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Disadvantages

1. Requires large area.
2. High construction cost and takes long time for erection.
3. Long transmission lines due to plant's location on hilly areas which are quite away from the load centre, leading to high cost in transmission.
4. Uncertainty about the availability of large amount of water due to dependence on weather conditions.

Example 5.

A hydroelectric generating plant is supplied from a reservoir of capacity $5 \times 10^6 \text{ m}^3$ at a head of 200 m. Find the total energy available in kWh if the overall efficiency is 75%.

Solution

$$\begin{aligned}\text{Weight of water available (W)} &= \text{Volume of water (V)} \times \text{density (S)} \times \\ &= 5 \times 10^6 \times 1000 \times 9.81 \\ &= 5 \times 10^9 \times 9.81 \\ &= 5 \times 10^9 \times 9.81 \text{ N} \\ &= 49,050 \text{ MN}\end{aligned}$$

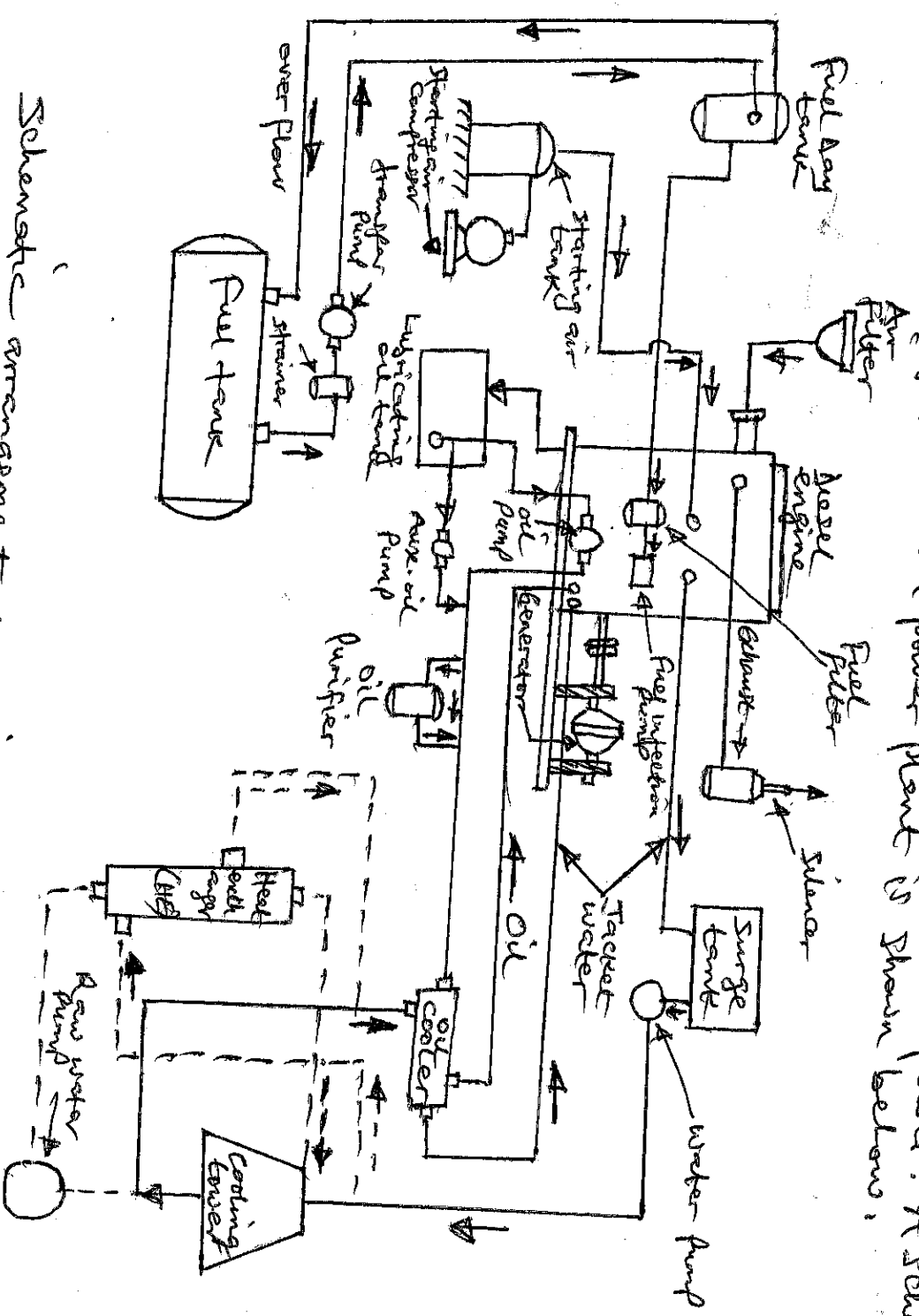
$$\begin{aligned}\text{Electrical energy available} &= \text{Weight of water} \times \text{head of water} \times \text{overall eff} \\ &= 49050 \times 10^6 \times 200 \times 0.75 \text{ watt sec.} \\ &= \frac{49050 \times 10^6 \times 200 \times 0.75}{3600 \times 1000} \text{ kWh}\end{aligned}$$

$$= \frac{7,357,500 \times 10^6}{3600000} = 2043,750 \text{ kWh}$$

$$= 2.044 \times 10^6 \text{ kWh}$$

Diesel Power Plant

A diesel power plant is a power plant in which a diesel engine is used as the prime mover for the generation of electrical energy. The diesel engine obtains energy from a liquid fuel usually called the diesel oil and converts it into Mechanical energy. An alternator or a dc generator mechanically coupled to the diesel engine converts the mechanical energy into electrical energy. As the generation cost is high due to high price of diesel, such power plants are only used to produce small power. A schematic arrangement of the diesel power plant is shown below.



Schematic arrangement of a Diesel power plant.

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Elements of Diesel Electric Power Plant

1. Diesel Engine:- This is the main component of the power plant that develops power. The diesel engines employed for diesel electric power plants may be four or two stroke engines. The engine may be horizontal or vertical type.
2. Fuel Supply System:- This consists of the storage tanks, strainers, fuel transfer pumps and connecting pipe work. The fuel oil is supplied to the plant site by rail or road. Fuel transfer pumps are required to transfer fuel from delivery point to storage tanks and from storage tanks to daily consumption tanks and then to the engine. Strainers are provided to remove suspended impurities and ensure that clean fuel is supplied to the engine by the fuel injection pump.
3. Air Intake System:- This includes the air filters and pipes that aid in supplying the necessary air to the engine manifold for fuel combustion. The filters are provided to remove dust particles from the air which may act as abrasive in the engine cylinder.
4. The Exhaust System:- This system includes the silencer and the connecting ducts. This system helps in discharging the engine exhaust gas outside the building and also to the atmosphere. The silencer is there to minimise the noise level.
5. Cooling System:- The heat released by the burning of fuel in the engine cylinder is partially converted into work. The remaining part of the heat passes through the cylinder walls, piston, rings etc. and may cause damage to the system. In order to keep the temperature of the engine part within the safe operating limits, cooling is provided. The system consists of a water source, pump and cooling towers. The water takes away the heat from the engine and itself becomes hot and then cooled by the cooling tower and is recirculated for cooling.
6. Lubricating System:- This system includes the lubricating oil pump, oil tanks, filters, coolers, purifiers and connecting pipes. The system reduces the wear of rubbing surfaces of the engine. The lubricating oil is drawn from the lubricating oil tank by the pump and is passed through filters to remove impurities. The clean lubricating oil is delivered to the points where

require lubrication. The oil coolers incorporated in the system keep the temperature of the oil low.

7. The engine starting system:— This includes the battery, starter, compressed air supply etc. This system is provided to rotate the engine initially, while starting, until the firing starts and the unit runs under its own power. Small sets are started manually by handles but for larger units, compressed air is used for starting.
8. Diesel Engine Governor (Attenuator):— They are directly coupled to the diesel engines.

Selection of Site for Diesel Electric Power Plants

The following factors should be considered while selecting a site for diesel electric power plant.

1. Distance from load Centre:— The site should be near the load centre as possible in order to avoid transmission cost and losses.
2. Availability of land:— The land should be available at cheap rate to keep the capital cost of the plant to the reasonable one.
3. Availability of fuel:— The fuel should be readily available and at a reasonable rate.
4. Transport:— The transportation facilities should be available.
5. Water Availability:— Water should be available in sufficient quantity for cooling purposes.
6. Distance from populated areas:— The site should be away from heavily populated areas because of noise and nuisance caused by exhaust.
7. Type of land:— The land should have good bearing capacity to withstand the load of the plant and also vibrations transmitted to the foundation ~~of the plant~~ and diesel engines.

Advantages of the Diesel Power Plants

1. The design and installation of the plant are simple.
2. It occupies less space as the number and size of the auxiliaries are small.
3. It can be located at any place.
4. It can be started quickly and can pick up load in a short time.
5. Requires less quantity of water for cooling.
6. Requires less operating staff.
7. The overall cost is much smaller than that of steam power plant of the same capacity.
8. The thermal efficiency of the plant is greater than that of a steam power plant.
9. Responds to varying loads with no difficulty.
10. No standby losses etc.

Disadvantages of the Diesel Power Plants

1. Operating cost is high due to high cost of diesel oil or fuel.
2. High maintenance and lubrication cost.
3. Can not be constructed in large size. Unit capacity is limited.
4. Does not work well under over load conditions for a long time.
5. Noise from the exhaust constitutes a nuisance problem.